# Data Science Toolbox Portfolio Questions

08 Algorithms for Data Science

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Block 8

## Portfolio 08

### Portfolio Question 8.1.P - Complexity Classes

- What is a complexity class, and how does it relate to the broader topic of "algorithmic complexity"?
  - What does $P \neq NP$ mean and is it important?
  - How does complexity relate to security and data science?
  - How does complexity relate to quantum computing?

### Portfolio Question 8.2.P - Hashing big data

- Imagine you are training an LLM and wanting to keep track of the documents its seen already, to avoid duplication and allow you to downweight topics by the proportion of computer generated content they contain.
  - How does the MinHash algorithm help solve this problem?
  - How would a sketch help?
  - Discuss the approach and any competition for the LLM problem.

### Portfolio Question 8.3.P - Empirical complexity

- You benchmark an algorithm whose theoretical complexity is $O(N \log N)$, but the log-log plot of run time against $N$ has a slope of 0 at small $N$, about 1 in the middle, and above 1 at large $N$.
  - What could cause each part of this curve?
  - When should you trust the theoretical complexity over the measurement, and when the reverse?

– How would you design a benchmark that you would trust to choose between two algorithms for a real deployment?